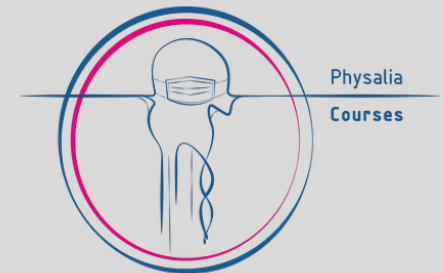


Quantification of gene expression

Epigenomics Data Analysis

Jacques Serizay

Physalia 2026



RNA-seq downstream analysis



Get .bcl files



Create fastq files



Or **bcl2fastq**



QC: remove/trim low quality reads

E.g. **cutadapt**



Align fastq to BAM

E.g. **bowtie2**, **STAR**



Filter duplicates, artifacts, ...

E.g. **samtools**



Generate tracks

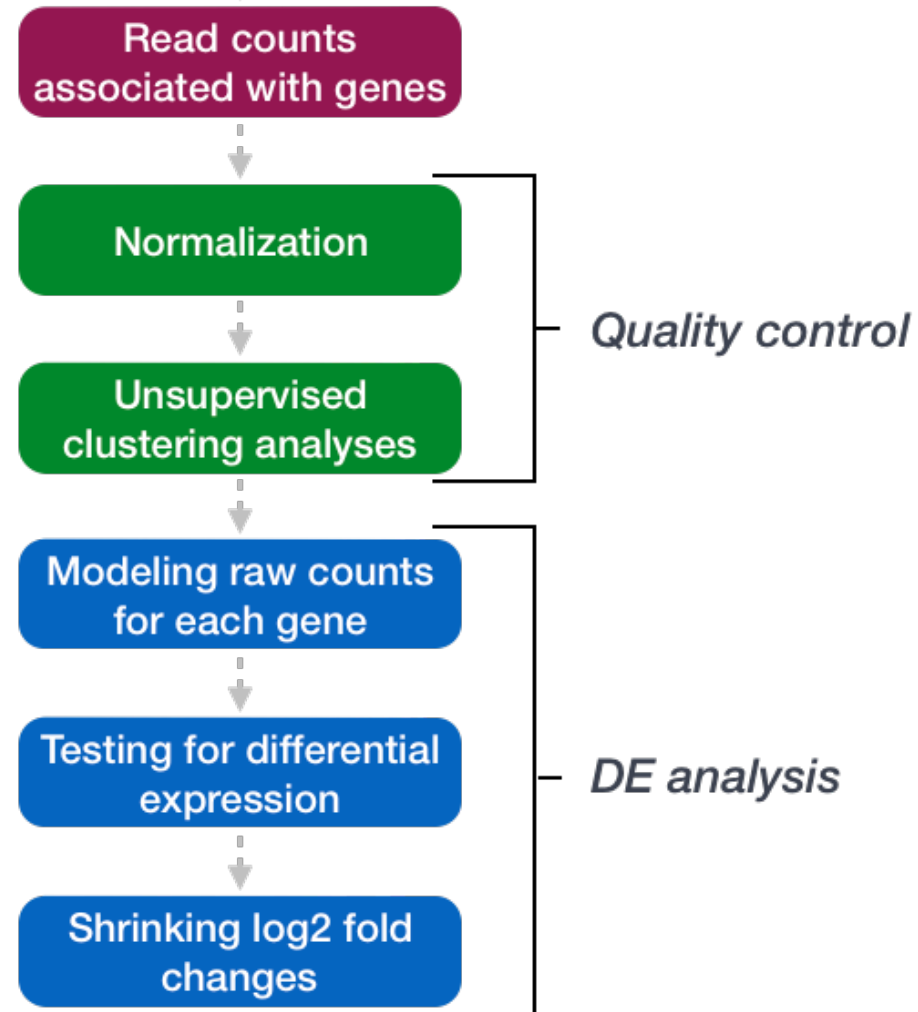
E.g. **deepTools**



Assay-specific downstream analysis

Transcript abundance quantification
Differential gene expression analysis
Gene ontology over-representation analysis

DESeq2 analysis workflow

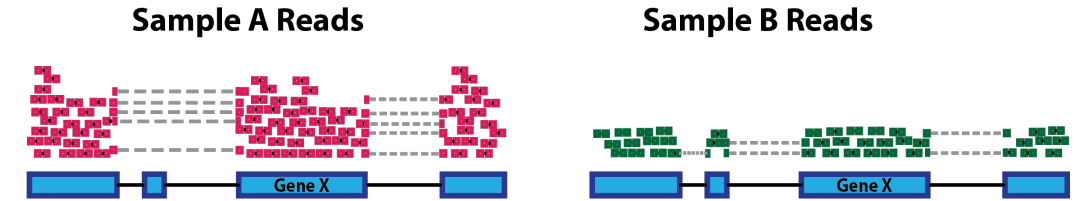


https://hbctraining.github.io/DGE_workshop/

Estimating gene/transcript abundance in RNA-seq

Important points when comparing gene abundance for different datasets/genes are:

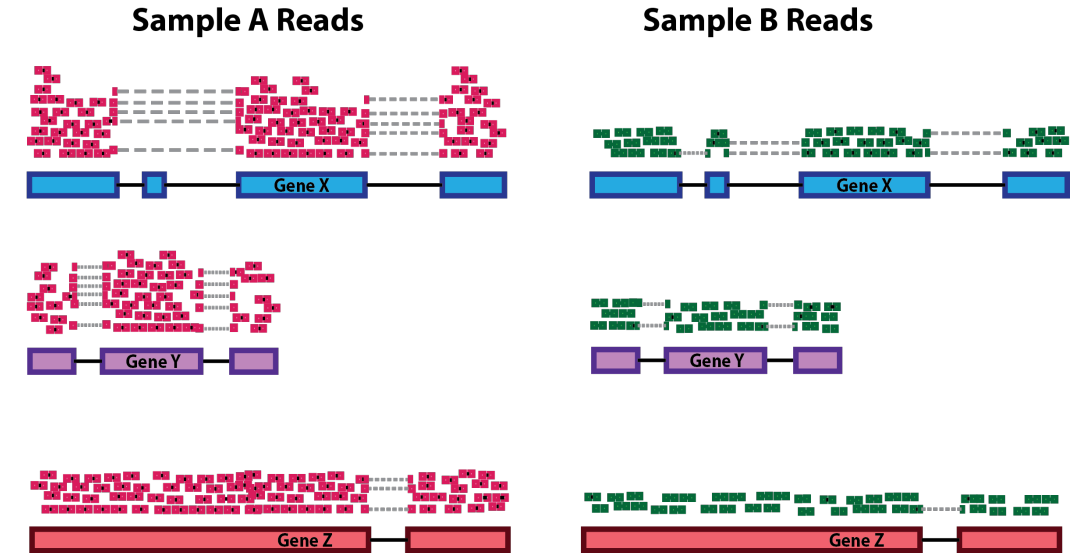
- Sequencing depth



Estimating gene/transcript abundance in RNA-seq

Important points when comparing gene abundance for different datasets/genes are:

- Sequencing depth
- Gene length

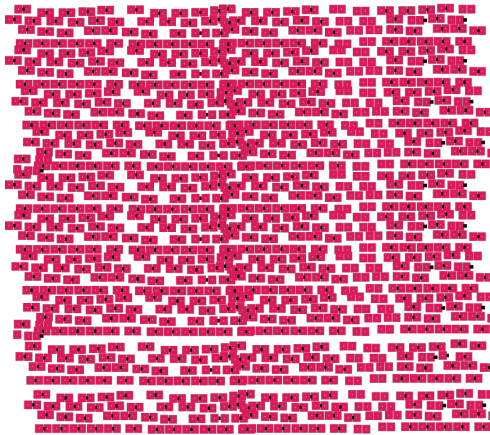
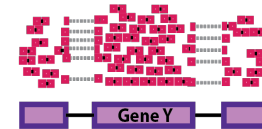
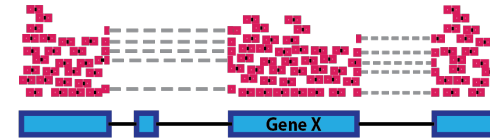


Estimating gene/transcript abundance in RNA-seq

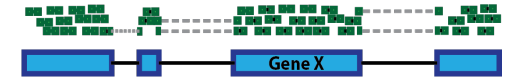
Important points when comparing gene abundance for different datasets/genes are:

- Sequencing depth
- Gene length
- RNA composition

Sample A Reads



Sample B Reads



https://hbctraining.github.io/DGE_workshop/

Estimating gene/transcript abundance in RNA-seq

- featureCounts can be used to count reads overlapping a set of gene annotations.

in bash

```
featureCounts \  
  -g gene_name \  
  -s 2 \  
  -p -B \  
  -T 16 \  
  -o data/counts/RNAseq_counts.tsv \  
  -a data/counts/hg38_Gencodev41.gtf \  
  data/mapping/RNA_ctl_1^hg38^filtered.bam \  
  data/mapping/RNA_ctl_2^hg38^filtered.bam \  
  data/mapping/RNA_ctl_3^hg38^filtered.bam \  
  data/mapping/RNA_foxj1_1^hg38^filtered.bam \  
  data/mapping/RNA_foxj1_2^hg38^filtered.bam
```

Estimating gene/transcript abundance in RNA-seq

- featureCounts can be used to count reads overlapping a set of gene annotations.
- Rsubread is an R package which wraps featureCounts function in R.

in bash

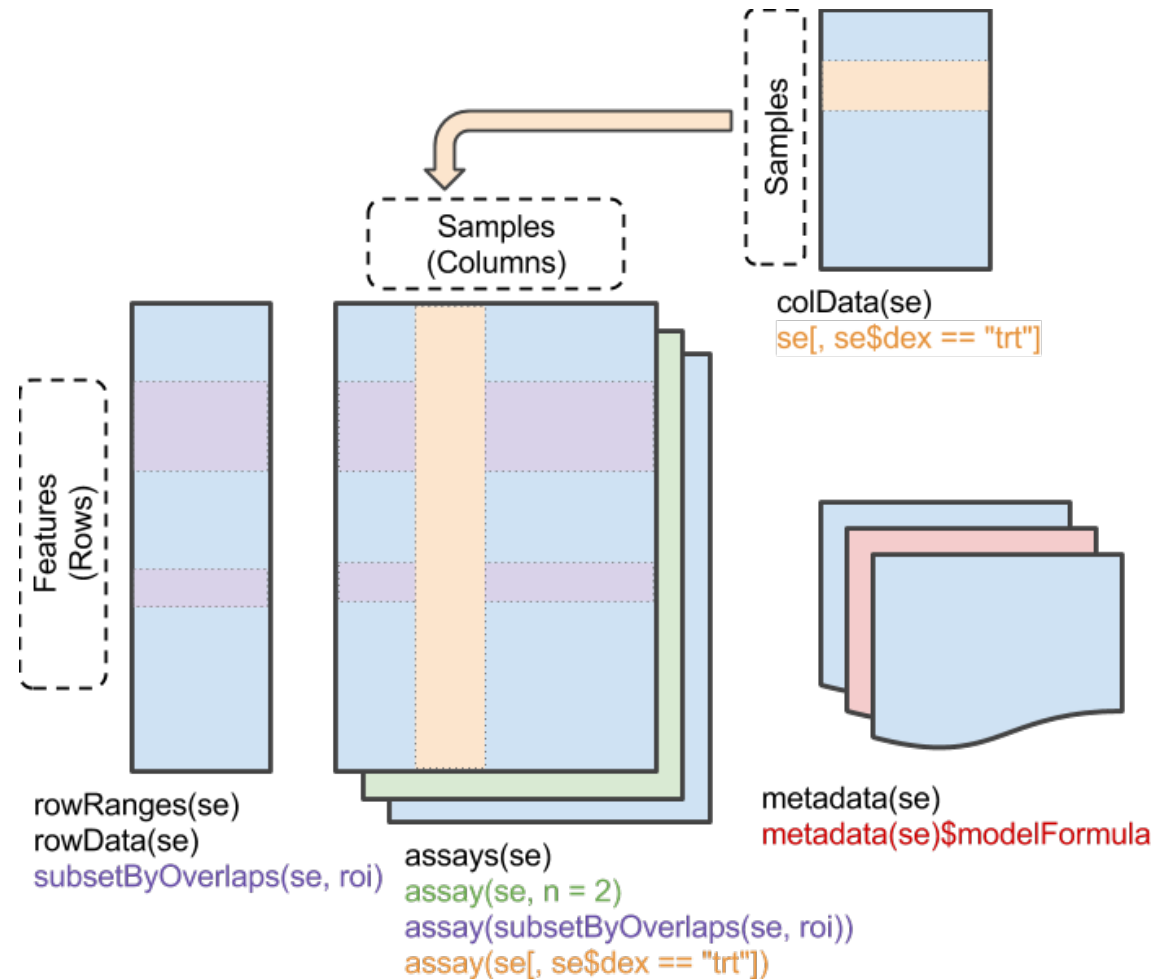
```
featureCounts \  
  -g gene_name \  
  -s 2 \  
  -p -B \  
  -T 16 \  
  -o data/counts/RNAseq_counts.tsv \  
  -a data/counts/hg38_Gencodev41.gtf \  
  data/mapping/RNA_ctl_1^hg38^filtered.bam \  
  data/mapping/RNA_ctl_2^hg38^filtered.bam \  
  data/mapping/RNA_ctl_3^hg38^filtered.bam \  
  data/mapping/RNA_foxj1_1^hg38^filtered.bam \  
  data/mapping/RNA_foxj1_2^hg38^filtered.bam
```

in R

```
cnts <- Rsubread::featureCounts(  
  files = c('...bam', '...bam'),  
  annot.ext = 'annots.gtf',  
  isGTFAnnotationFile = TRUE,  
  GTF.featureType = 'sequence_feature',  
  GTF.attrType = 'id',  
  isPairedEnd = TRUE,  
  nthreads = 16  
)
```


DESeq2 analysis workflow

- DESeq2 is based on a SummarizedExperiment object.



SummarizedExperiment

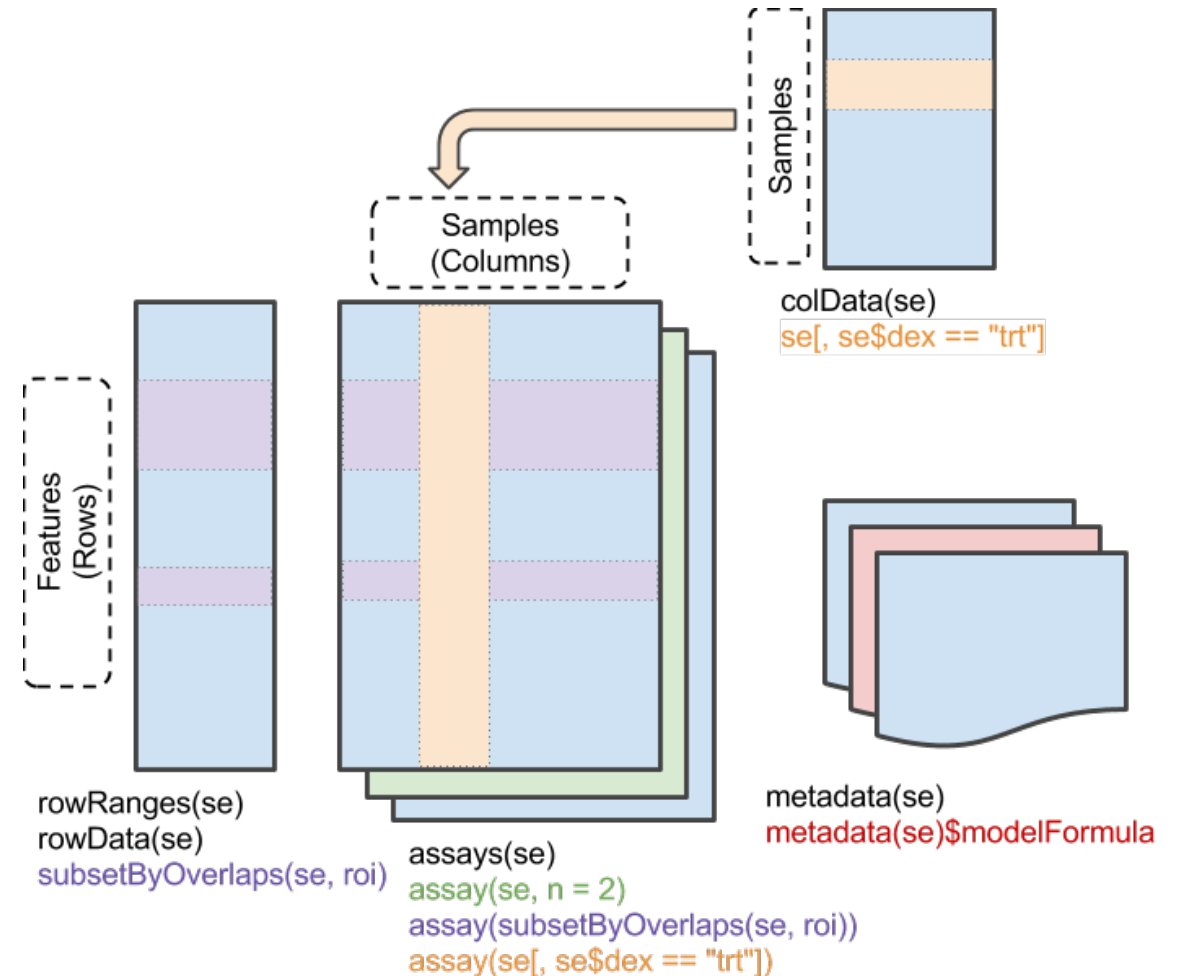
Published: 29 January 2015

Orchestrating high-throughput genomic analysis with Bioconductor

Wolfgang Huber , Vincent J Carey, Robert Gentleman, Simon Anders, Marc Carlson, Benilton S Carvalho, Hector Corrada Bravo, Sean Davis, Laurent Gatto, Thomas Girke, Raphael Gottardo, Florian Hahne, Kasper D Hansen, Rafael A Irizarry, Michael Lawrence, Michael I Love, James MacDonald, Valerie Obenchain, Andrzej K Oleś, Hervé Pagès, Alejandro Reyes, Paul Shannon, Gordon K Smyth, Dan Tenenbaum, Levi Waldron & Martin Morgan [-Show fewer authors](#)

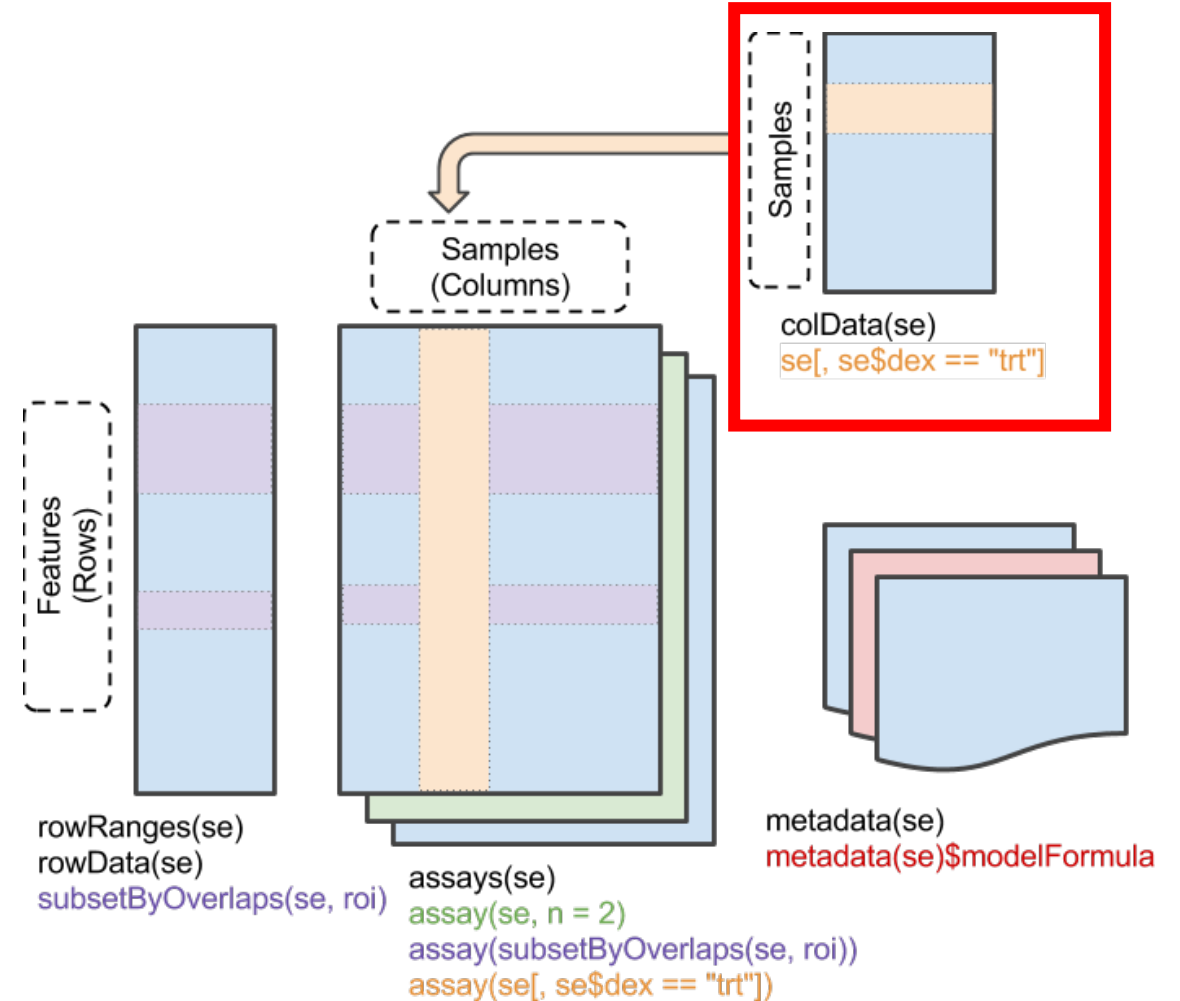
Nature Methods **12**, 115–121(2015) | [Cite this article](#)

Huber et al., Nat. Meth. 2015



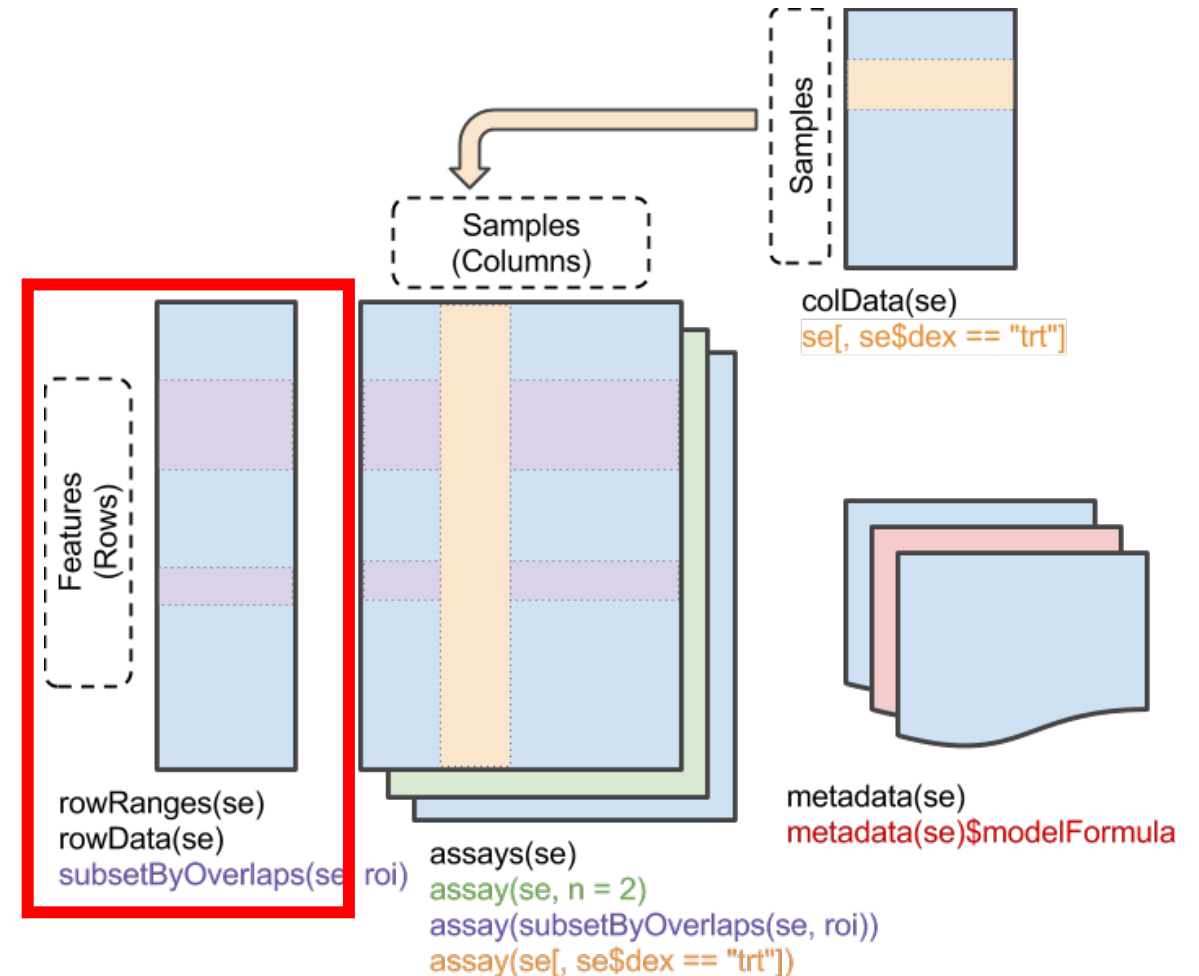
SummarizedExperiment

- `colData()`: Annotations on each column, as a DataFrame.
 - E.g., description of each sample



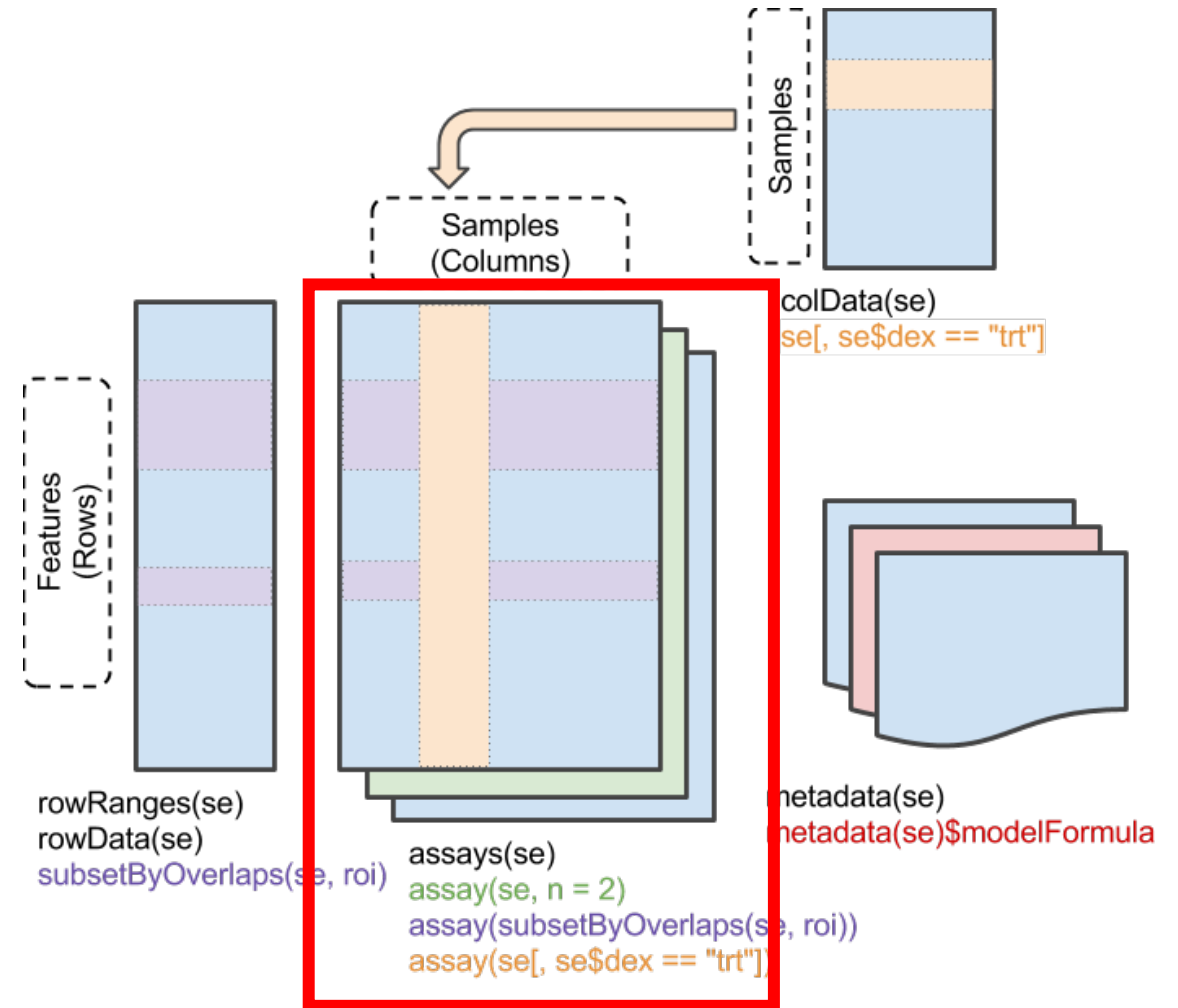
SummarizedExperiment

- `colData()`: Annotations on each column, as a DataFrame.
 - E.g., description of each sample
- `rowData/rowRanges()`: Annotations on each row.
 - E.g., coordinates of gene / exons / peaks in transcripts / etc.



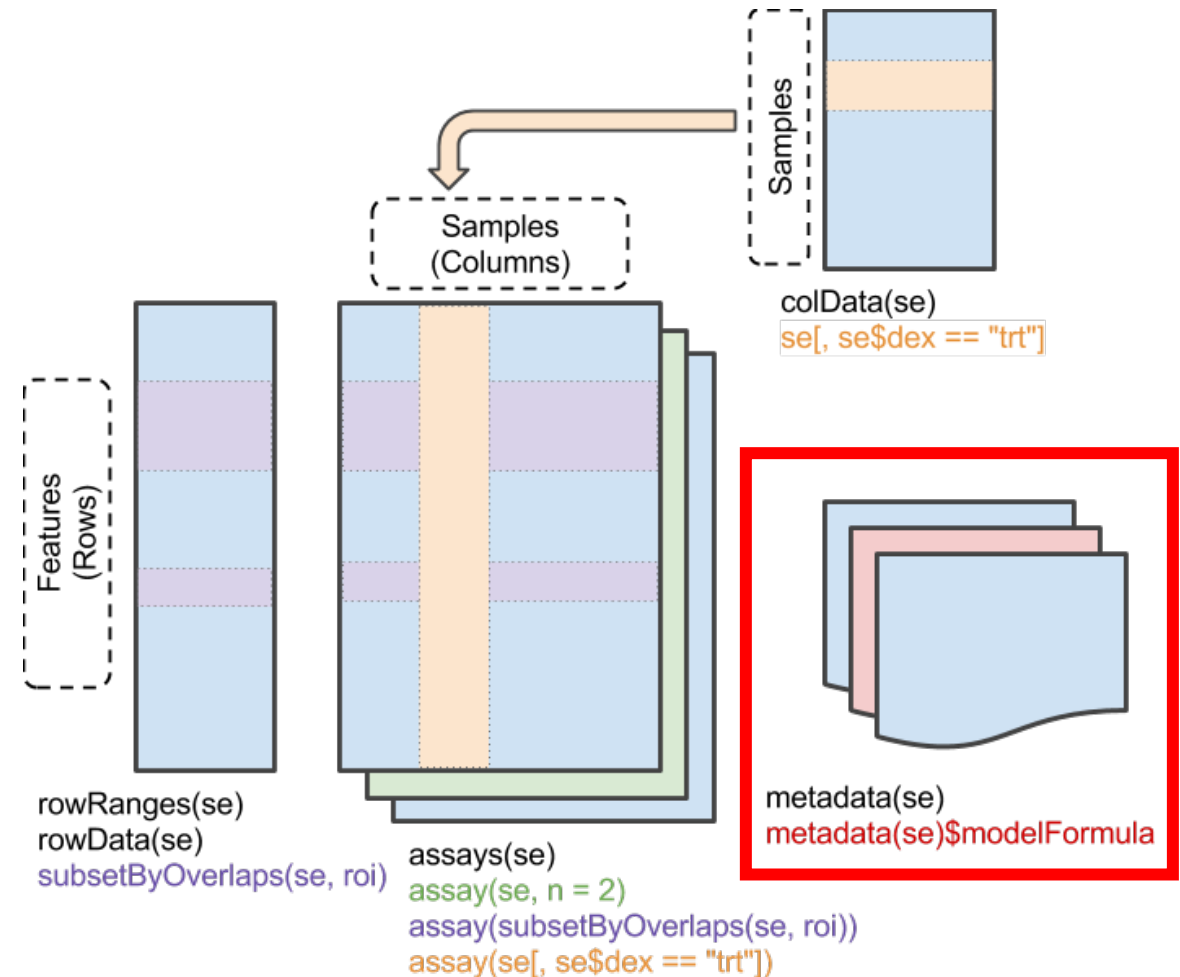
SummarizedExperiment

- `colData()`: Annotations on each column, as a DataFrame.
 - E.g., description of each sample
- `rowData/rowRanges()`: Annotations on each row.
 - E.g., coordinates of gene / exons / peaks in transcripts / etc.
- `assay()`, `assays()`: A matrix-like or list of matrix-like objects of identical dimension
 - rows: refer to **rowRanges**: genes, genomic coordinates, etc.
 - columns: refer to **colData**: samples, cells, etc.
 - Implements `dim()`, `dimnames()` and 2-dimensional [,]
 - **Can be several assays!!!**



SummarizedExperiment

- `colData()`: Annotations on each column, as a DataFrame.
 - E.g., description of each sample
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- `assay()`, `assays()`: A matrix-like or list of matrix-like objects of identical dimension
 - rows: refer to **rowRanges**: genes, genomic coordinates, etc.
 - columns: refer to **colData**: samples, cells, etc.
 - Implements `dim()`, `dimnames()` and 2-dimensional [,]
 - **Can be several assays!!!**
- `metadata()`: List of unstructured metadata describing the overall content of the object.



SummarizedExperiment

SummarizedExperiment is a key data structure in **Bioconductor**, reused by or modified in:

- clusterExperiment
- SingleCellExperiment
- LoomExperiment
- MsExperiment
- TreeSummarizedExperiment
- SpatialExperiment
- And 100s more...

Depends On Me

[AffiXcan](#), [airway](#), [alabaster.se](#), [AllelicImbalance](#), [ASpediaFI](#), [atena](#), [bambu](#), [BDMMAcorrect](#), [benchmarkfdrData2019](#), [BiocSlearn](#), [BioPlex](#), [BiSeq](#), [bnbc](#), [bodymapRat](#), [bsseq](#), [CAGEfightR](#), [celaref](#), [celldex](#), [clusterExperiment](#), [compartmap](#), [CoreGx](#), [coseq](#), [csaw](#), [CSSQ](#), [curatedAdipoChIP](#), [curatedAdipoRNA](#), [curatedMetagenomicData](#), [DaMiRseq](#), [deco](#), [deepSNV](#), [DeMixT](#), [DESeq2](#), [DEXseq](#), [DiffBind](#), [diffcoexp](#), [diffHic](#), [divergence](#), [DMCFB](#), [DMCHMM](#), [ENmix](#), [EnrichmentBrowser](#), [epigenomix](#), [evaluomeR](#), [EventPointer](#), [exomePeak2](#), [ExperimentSubset](#), [ExpressionAtlas](#), [extraChIPs](#), [FEAST](#), [fission](#), [FRASER](#), [GenomicAlignments](#), [GenomicFiles](#), [GenomicSuperSignature](#), [GRmetrics](#), [GSEABenchmarkR](#), [HDCytoData](#), [HelloRanges](#), [hermes](#), [HiCDOC](#), [HighlyReplicatedRNASeq](#), [hipathia](#), [HMP16SDData](#), [IntAD](#), [InteractionSet](#), [InterEst](#), [ISEE](#), [ISEEhex](#), [ISEEhub](#), [ISLET](#), [isomiRs](#), [ivygapSE](#), [lefser](#), [lipidr](#), [LoomExperiment](#), [Macarron](#), [made4](#), [MatrixQCvis](#), [MBASED](#), [MetaGxOvarian](#), [MetaGxPancreas](#), [methrix](#), [methylPipe](#), [MethylSeqData](#), [MetNet](#), [mia](#), [miaViz](#), [MicrobiomeBenchmarkData](#), [microbiomeDataSets](#), [microRNAome](#), [minfi](#), [miRmine](#), [moanin](#), [MouseGastrulationData](#), [MouseThymusAgeing](#), [mpa](#), [MultiAssayExperiment](#), [NADfinder](#), [NBAMSeq](#), [NewWave](#), [ObMITI](#), [OUTRIDER](#), [padma](#), [parathyroidSE](#), [PDATK](#), [phenomis](#), [PhiPData](#), [profileplyr](#), [qmtools](#), [qsvaR](#), [recount](#), [recount3](#), [RegEnrich](#), [REMP](#), [restfulSE](#), [restfulSEData](#), [ROCpAI](#), [rqt](#), [runibic](#), [sampleClassifierData](#), [Scale4C](#), [scAnnotatR](#), [scGPS](#), [scMultiome](#), [scone](#), [screenCounter](#), [scTreeViz](#), [SDAMS](#), [SeqGate](#), [SEtools](#), [SGSeq](#), [signatureSearch](#), [SingleCellExperiment](#), [singleCellITK](#), [SingleR](#), [soGGi](#), [spatialDmexsim](#), [sqpn](#), [sqpnData](#), [sRACIPE](#), [ssPATHS](#), [stageR](#), [SummarizedBenchmark](#), [survtype](#), [TENxIO](#), [tidySummarizedExperiment](#), [timecoursedata](#), [TissueEnrich](#), [TNBC.CMS](#), [TREG](#), [tuberculosis](#), [UMI4Cats](#), [VanillaICE](#), [VariantAnnotation](#), [VariantExperiment](#), [velociraptor](#), [weitrax](#), [yamss](#), [zinbwave](#)

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[tidybulk](#), [tidySingleCellExperiment](#), [TOAST](#), [tomoda](#), [ToxicoGx](#), [tradeSeq](#), [TrajectoryUtils](#), [transformGamPoi](#), [traviz](#), [TreeSummarizedExperiment](#), [Trendy](#), [tricycle](#), [TSCAN](#), [tsr](#), [TTMap](#), [TVTB](#), [tximeta](#), [UCell](#), [VAExprs](#), [VariantFiltering](#), [VDJdive](#), [vidger](#), [Voyager](#), [wpm](#), [xcms](#), [zellkonverter](#), [zFPKM](#)



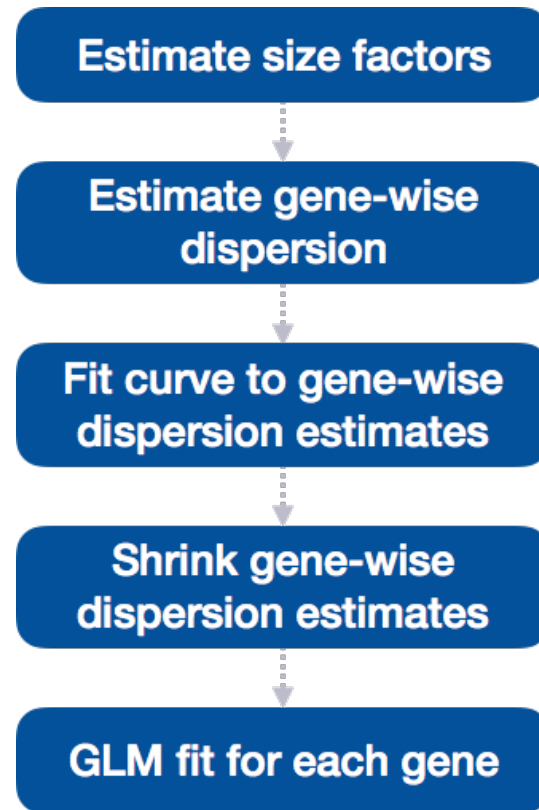
DESeq2 analysis workflow

- DESeq2 is based on a SummarizedExperiment object.
- A **formula** needs to be specified

```
> dds <- DESeqDataSet(counts, design = ~ timepoint)
```


DESeq2 analysis workflow

- Differential expression analysis is as simple as `DESeq()`



	sample2	WT	2	No drug	
	sample3	KO	1	Drug	
• DE	sample4	KO	2	drug	
	sample5	WT	1	drug	
• Af	sample6	WT	2	drug	

- Differential expression analysis is as simple as DESeq ()

```
> dds <- DESeqDataSet(counts, design = ~ Condition + Treatment)
> dds <- DESeq(dds)
estimating size factors
estimating dispersions
gene-wise dispersion estimates
mean-dispersion relationship
final dispersion estimates
fitting model and testing
> dds
class: DESeqDataSet
dim: 6513 10
metadata(1): version
assays(4): counts mu H cooks
rownames: NULL
rowData names(37): summit peakID ... deviance maxCooks
colnames: NULL
colData names(5): sample timepoint replicate bam sizeFactor
```

DESeq2 analysis workflow

4) Extract results

- ❖ We can use `DESeq2::results()`
- ❖ DESeq2 extracts results for pairwise comparison between 2 conditions

```
> contrasts
  15_v_00  30_v_00  45_v_00  60_v_00  30_v_15  45_v_15  60_v_15  45_v_30  60_v_30  60_v_45
[1,] "timepoint" "timepoint" "timepoint" "timepoint" "timepoint" "timepoint" "timepoint" "timepoint" "timepoint" "timepoint"
[2,] "15"        "30"        "45"        "60"        "30"        "45"        "60"        "45"        "60"        "60"
[3,] "00"        "00"        "00"        "00"        "15"        "15"        "15"        "30"        "30"        "45"
```

DESeq2 analysis workflow

```
> contrasts
      15_v_00  30_v_00  45_v_00  60_v_00  30_v_15  45_v_15  60_v_15  45_v_30  60_v_30  60_v_45
[1,] "timepoint" "timepoint" "timepoint" "timepoint" "timepoint" "timepoint" "timepoint" "timepoint" "timepoint" "timepoint"
[2,] "15"        "30"        "45"        "60"        "30"        "45"        "60"        "45"        "60"        "60"
[3,] "00"        "00"        "00"        "00"        "15"        "15"        "15"        "30"        "30"        "45"
```

```
> results(dds, contrast = contrasts[, 2])
log2 fold change (MLE): timepoint 30 vs 00
Wald test p-value: timepoint 30 vs 00
DataFrame with 6513 rows and 6 columns
```

	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
1	1643.7457	-0.900131	0.225114	-3.99856	6.37297e-05	0.002809493
2	106.1707	-0.876365	0.313066	-2.79930	5.12139e-03	0.066589377
3	134.4141	-0.753498	0.311437	-2.41942	1.55451e-02	0.133487340
4	203.8053	-1.094019	0.227780	-4.80296	1.56338e-06	0.000140911
5	67.2922	-1.001061	0.336647	-2.97362	2.94309e-03	0.046569804
...	...	...	...	...	...	...
6509	36.1437	-0.0313828	0.417826	-0.0751098	0.9401274	0.979001
6510	603.5468	-0.4020281	0.181778	-2.2116386	0.0269916	0.185475
6511	42.8792	-0.3776449	0.378523	-0.9976807	0.3184342	0.633629
6512	15.2541	-0.5818438	0.642764	-0.9052221	0.3653477	NA
6513	37.8941	-0.8396196	0.427293	-1.9649731	0.0494173	0.258224

DESeq2 analysis workflow

```
> contrasts
  15_v_00 30_v_00 45_v_00 60_v_00 30_v_15 45_v_15 60_v_15 45_v_30 60_v_30 60_v_45
[1,] "timepoint" "timepoint" "timepoint" "timepoint" "timepoint" "timepoint" "timepoint" "timepoint" "timepoint" "timepoint"
[2,] "15" "30" "45" "60" "30" "45" "60" "45" "60" "60"
[3,] "00" "00" "00" "00" "15" "15" "15" "30" "30" "45"
```

```
> results(dds, contrast = contrasts[, 2])
```

log2 fold change (MLE): timepoint 30 vs 00

Wald test p-value: timepoint 30 vs 00

DataFrame with 6513 rows and 6 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
1	1643.7457	-0.900131	0.225114	-3.99856	6.37297e-05	0.002809493
2	106.1707	-0.876365	0.313066	-2.79930	5.12139e-03	0.066589377
3	134.4141	-0.753498	0.311437	-2.41942	1.55451e-02	0.133487340
4	203.8053	-1.094019	0.227780	-4.80296	1.56338e-06	0.000140911
5	67.2922	-1.001061	0.336647	-2.97362	2.94309e-03	0.046569804
...	...	...	...	...	...	...
6509	36.1437	-0.0313828	0.417826	-0.0751098	0.9401274	0.979001
6510	603.5468	-0.4020281	0.181778	-2.2116386	0.0269916	0.185475
6511	42.8792	-0.3776449	0.378523	-0.9976807	0.3184342	0.633629
6512	15.2541	-0.5818438	0.642764	-0.9052221	0.3653477	NA
6513	37.8941	-0.8396196	0.427293	-1.9649731	0.0494173	0.258224

DESeq2 analysis workflow

```
> contrasts
  15_v_00  30_v_00  45_v_00  60_v_00  30_v_15  45_v_15  60_v_15  45_v_30  60_v_30  60_v_45
[1,] "timepoint" "timepoint" "timepoint" "timepoint" "timepoint" "timepoint" "timepoint" "timepoint" "timepoint" "timepoint"
[2,] "15"        "30"        "45"        "60"        "30"        "45"        "60"        "45"        "60"        "60"
[3,] "00"        "00"        "00"        "00"        "15"        "15"        "15"        "30"        "30"        "45"
```

```
> results(dds, contrast = contrasts[, 2])
log2 fold change (MLE): timepoint 30 vs 00
Wald test p-value: timepoint 30 vs 00
DataFrame with 6513 rows and 6 columns
```

	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
1	1643.7457	-0.900131	0.225114	-3.99856	6.37297e-05	0.002809493
2	106.1707	-0.876365	0.313066	-2.79930	5.12139e-03	0.066589377
3	134.4141	-0.753498	0.311437	-2.41942	1.55451e-02	0.133487340
4	203.8053	-1.094019	0.227780	-4.80296	1.56338e-06	0.000140911
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...	...	...	...	...	...	...
6509	36.1437	-0.0313828	0.417826	-0.0751098	0.9401274	0.979001
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DESeq2 analysis workflow

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[1,] "timepoint" "timepoint" "timepoint" "timepoint" "timepoint" "timepoint" "timepoint" "timepoint" "timepoint" "timepoint"
[2,] "15"        "30"        "45"        "60"        "30"        "45"        "60"        "45"        "60"        "60"
[3,] "00"        "00"        "00"        "00"        "15"        "15"        "15"        "30"        "30"        "45"
```

```
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...	...	...	...	...	...	...
6509	36.1437	-0.0313828	0.417826	-0.0751098	0.9401274	0.979001
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6513	37.8941	-0.8396196	0.427293	-1.9649731	0.0494173	0.258224

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- Rlog transforms the count data to the **log2 scale** in a way which
 - 1) minimizes differences between samples for rows with small counts
 - 2) normalizes with respect to library size

DESeq2::rlog()